

André Carlon

Curriculum Vitae

Department of Mathematics, RWTH-Aachen
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Current position

2024– **Postdoctoral Researcher, Chair of Mathematics for Uncertainty Quantification (Prof. Raúl Tempone)**, RWTH Aachen University, Aachen, Germany.
Research on stochastic optimization and Bayesian uncertainty quantification.

Education

2015–2019 **Ph.D. in Structural Engineering**, UFSC, Florianópolis, Brazil.
2013–2015 **M.Sc. in Structural Optimization (Engineering)**, UFSC, Florianópolis, Brazil.
2007–2013 **B.Sc. in Civil Engineering**, UFSC, Florianópolis, Brazil.

Doctorate Dissertation

Title *Bayesian optimization of experiments using stochastic gradient methods*
Supervisor Professor Rafael Holdorf Lopez

Masters Thesis

Title *Development of a new operator for metaheuristic algorithms based on stiffness maximization and its application to the optimization of truss structures*
Supervisors Professor Rafael Holdorf Lopez & Professor Leandro Flackel Miguel

Experience

2020–2024 **Post-doctoral fellow**, Stochastic Numerics Research Group, KAUST, Thuwal, Saudi Arabia.
2017–2018 **Visiting student**, King Abdullah University of Science and Technology (KAUST), Thuwal, Saudi Arabia.
Worked with Prof. Raúl Tempone on optimal experimental design using stochastic gradient methods.

Grants and Awards

- 2025–2027 **Development of a methodology based on digital twins and artificial intelligence to support decision-making in the maintenance and management of railway bridges**, *Funding: CNPq/MCTI/FNDCT*, Amount: R\$ 360,000 (€ 57,000 approx.), Role: Scientific contributor (methodology lead); responsibilities in stochastic optimization for sensor placement and training of digital-twin models.
- 2022–2025 **KAUST Collaborative Research Grant**, *Funding: KAUST*, Amount: \$ 900,000 (approx.), Role: Contributor (text drafting and preliminary results).

Student supervision

Coordinator of the B.Sc./M.Sc. seminar “Data Science under Uncertainty” (RWTH Aachen, 2024–2025), supervising students on uncertainty quantification topics.

Collaborations and partnerships

- 2025–2027 MTI – Manusafe project: Uncertainty quantification for surface integrity monitoring in machining processes (e.g., turning and milling).
- 2025–2026 German Aerospace Center (DLR) – Optimization of UAV paths for pollutant source localization.
- 2025–2027 CNPq/MCTI/FNDCT: Development of a methodology based on digital twins and artificial intelligence to support decision-making in the maintenance and management of railway bridges.

Teaching experience

Coordinator, “Data Science under Uncertainty” (seminar), RWTH–Aachen, 2024–2025 (BSc).

Co-instructor, “Numerical Methods for Random Partial Differential Equations”, KAUST (2020–2023), RWTH–Aachen (2024) (PhD/MSc).

Co-instructor, “Stochastic Numerics with Applications in Simulation and Data Science”, KAUST (2020–2024) (PhD/MSc).

Service activities

Associate Editor, “Statistics and Computing” (Q1), since 2025.

Organizer, “Stochastic Numerics and Statistical Learning” workshop, five editions, 2022–2026.

Regular reviewer for Computer Methods in Applied Mechanics and Engineering; Reliability Engineering & System Safety; and related venues.

Skills

- Core Bayesian inference; Uncertainty quantification; Stochastic and deterministic optimization; Reliability and structural engineering; Convex and variational analysis; Numerical methods for PDEs
- Coding Python (JAX, NumPy, Optax), MATLAB, Julia, $\text{\LaTeX}$
- Other Elasticity and non-linear structural analysis; Scientific machine learning; Open-source software development

Languages

- Portuguese Native language
- English Fluent
- Spanish Intermediate
- German Beginner

Publications

- 1 **Carlon, A.G.**, Espath, L., Holdorf, R., Tempone, R. (2026). Multi-iteration stochastic optimizers. *Applied Mathematics & Optimization*, 93(3).
- 2 **Carlon, A.G.**, Espath, L., Tempone, R. (2025). Efficient Stochastic BFGS methods Inspired by Bayesian Principles. *arXiv preprint* arXiv:2507.07729.
- 3 **Carlon, A.G.**, Espath, L., Tempone, R. (2024). Approximating Hessian matrices using Bayesian inference: a new approach for quasi-Newton methods in stochastic optimization. *Optimization Methods and Software*, 39(6), 1352–1382.
- 4 Bartuska, A., **Carlon, A.G.**, Espath, L., Krumscheid, S., Tempone, R. (2024). Multilevel randomized quasi-Monte Carlo estimator for nested integration. *arXiv:2412.07723*.
- 5 **Carlon, A.G.**, Maia, C.D.C.D., Lopez, R.H., Torii, A.J., Miguel, L.F.F. (2023). A polynomial chaos efficient global optimization approach for Bayesian optimal experimental design. *Probabilistic Engineering Mechanics*, 72, 103454.
- 6 Bartuska, A., **Carlon, A.G.**, Espath, L., Krumscheid, S., Tempone, R. (2023). Double-loop randomized quasi-Monte Carlo estimator for nested integration. *arXiv:2302.14119*.
- 7 **Carlon, A.G.**, Kroetz, H.M., Torii, A.J., Lopez, R.H., Miguel, L.F.F. (2022). Risk optimization using the Chernoff bound and stochastic gradient descent. *Reliability Engineering & System Safety*, 223, 108512.
- 8 **Carlon, A.G.**, Dia, B.M., Espath, L., Lopez, R.H., Tempone, R. (2020). Nesterov-aided stochastic gradient methods using Laplace approximation for Bayesian design optimization. *Computer Methods in Applied Mechanics and Engineering*, 363, 112909.
- 9 **Carlon, A.G.**, Lopez, R.H., Espath, L., Miguel, L.F.F., Beck, A.T. (2019). A stochastic gradient approach for the reliability maximization of passively controlled structures. *Engineering Structures*, 186, 1–12.

- 10 Souza Cursi, J.E., Lopez, R.H., **Carlon, A.G.** (2019). A new approach for state estimation. Chapter in *Uncertainty Modeling for Engineering Applications*.
- 11 Lopez, R.H., Cursi, J.E.S., **Carlon, A.G.** (2018). A state estimation approach based on stochastic expansions. *Computational and Applied Mathematics*, 37, 3399–3430.

Monographs

Carlon, A.G. (2019). Bayesian optimization of experiments using stochastic gradient methods. *Doctoral dissertation*, UFSC.

Carlon, A.G. (2015). Development of a new operator for metaheuristic algorithms based on stiffness maximization and its application to the optimization of truss structures. *Masters dissertation*, UFSC.

Carlon, A.G. (2013). Optimization of steel truss structures applying the ICA algorithm. *Undergraduate thesis*, UFSC.

Conferences

- 1 **Carlon, A.G.**, Espath, L., Tempone, R. A Bayesian Stochastic L-BFGS. Stochastic Numerics and Inverse Problems in Sweden (SNIPS), 2025.
- 2 **Carlon, A.G.**, Espath, L., Tempone, R. Bayesian Hessian Approximation for Robust Quasi-Newton Methods in Stochastic Optimization. Modeling and Optimization: Theory and Applications (MOPTA) Conference, 2025.
- 3 **Carlon, A.G.**, Beck, J., Tempone, R. Optimal Bayesian experimental design using stochastic gradient descent with adaptive sample sizes. ADMOS 2025, keynote presentation.
- 4 **Carlon, A.G.**, Beck, J., Tempone, R. Adaptive double-loop Monte Carlo gradient estimators for Bayesian optimal experimental design. 5th International Conference on Uncertainty Quantification in Computational Science and Engineering, 2023.
- 5 **Carlon, A.G.**, Kroetz, H.M., Lopez, R.H., Miguel, L.F.F. A stochastic gradient descent approach for risk optimization using the Chernoff bound. XLIII Ibero-Latin-American Congress on Computational Methods in Engineering, ABMEC, 2022.
- 6 **Carlon, A.G.**, Beck, J., Tempone, R. Adaptive stochastic gradient descent for Bayesian optimal experimental design. 15th International Conference on Monte Carlo and Quasi-Monte Carlo Methods in Scientific Computing, 2022.
- 7 **Carlon, A.G.**, Dia, B.M., Espath, L., Lopez, R.H., Tempone, R. Maximization of the Expected Information Gain of Experiments Using Stochastic Gradient Descent. 16th U.S. National Congress on Computational Mechanics, 2021.
- 8 **Carlon, A.G.**, Dia, B.M., Espath, L., Lopez, R.H., Tempone, R. Bayesian optimization of experiments using stochastic gradient descent and Nesterov acceleration. 6th ECCOMAS Young Investigators Conference (YIC2021), 2021.

- 9 **Carlon, A.G.**, Dia, B.M., Espath, L., Lopez, R.H., Tempone, R. Optimization of experimental design for non-linear models using stochastic gradient descent. 42nd Ibero-Latin-American Congress on Computational Methods in Engineering and 3rd Pan American Congress on Computational Mechanics, 2021.
- 10 **Carlon, A.G.**, Dia, B.M., Espath, L., Lopez, R.H., Tempone, R. Stochastic optimization for Bayesian design of experiments. 14th International Conference on Monte Carlo & Quasi-Monte Carlo Methods in Scientific Computing, 2020.
- 11 **Carlon, A.G.**, Torii, A.J., Lopez, R.H., Souza de Cursi, J.E. Stochastic Gradient Descent for Risk Optimization. Proceedings of the 5th International Symposium on Uncertainty Quantification and Stochastic Modelling, 2020.
- 12 **Carlon, A.G.**, Lopez, R.H., Espath, L., Miguel, L.F.F. Optimal Bayesian Design for Electrical Impedance Tomography Experiments. Proceedings of the joint ICVRAM ISUMA UNCERTAINTIES conference, 2018.
- 13 **Carlon, A.G.**, Lopez, R.H., Espath, L. Stochastic optimization for design of experiments. Proceedings of the XXXVIII Iberian Latin-American Congress on Computational Methods in Engineering, 2017.
- 14 **Carlon, A.G.**, Lopez, R.H., Espath, L., Miguel, L.F.F. A Random Perturbation of the Gradient approach for Global Optimization of Stochastic Functions. Proceedings of the European Conference on Stochastic Optimization, 2017.
- 15 Lopez, R.H., Souza de Cursi, J.E., **Carlon, A.G.**. State estimation based on stochastic polynomials and variational approximation. Uncertainties 2016.
- 16 **Carlon, A.G.**, Lopez, R.H., Souza de Cursi, J.E., Miguel, L.F.F. A collocation approach for state estimation using polynomial chaos expansions. XXXVII Iberian Latin-American Congress on Computational Methods in Engineering, 2016.
- 17 Lopez, R.H., Souza de Cursi, J.E., **Carlon, A.G.**. New approach for the state estimation problem based on the representation of random variables using stochastic polynomials. 3rd International Symposium on Uncertainty Quantification and Stochastic Modeling, 2016.